LAGOS BUSINESS SCHOOL · MEMBA 14 · EEGB · PART I

SESSION 4

Market Structure, Platforms & Competition Policy



From markets, to macro, to decisions that matter — in Nigeria and the world.

Reading: Fintech and Finance Transformation: The Rise of Ant Financial Services (917-0029-1) · Facilitator: Kelechi Nnamdi

RECAP

Session 3 in Three Ideas



Trend × season

Read demand as trend, seasonality and shocks — then model each part.



Backtest and range

Judge forecasts on data they have not seen; give a range and scenarios.



Size two ways

Top-down and bottom-up market sizing should roughly agree.

TODAY'S PLAN

Session 4 at a Glance — 75 Minutes

How many rivals you face — and whether a platform's size helps or hurts its customers.



Recap & Hook

Is the biggest payments app too big?

Market Structure

From perfect competition to monopoly

Platforms

Two-sided markets, network effects, switching costs, data

Competition Policy & Lab

Measuring concentration; the regulator's view

BY THE END OF TODAY

Session 4 Learning Objectives

You will be able to:

01

Classify a Market

Place an industry on the spectrum from perfect competition to monopoly, and say what that means for prices and profits.

02

Understand Platforms

Explain two-sided markets, network effects, switching costs and data as sources of market power.

03

Think Like a Regulator

Measure concentration (CR and HHI), and judge when a dominant firm or a merger becomes a competition problem.

THE HOOK

Is the Biggest App Too Big?



“Five years ago nobody had heard of them. Now every market woman in Lagos has their POS — and their fees just went up.”

Size alone is not a crime. A competition economist asks:

How did it get big?

By being better and cheaper (good), or by locking customers in and shutting rivals out (bad)?

Can rivals still enter?

If a new app can win users and merchants, the leader's power is limited.

What happens to prices?

Rising fees, falling quality or slower innovation are the warning signs.

NIGERIAN BUSINESS CASE

Nigeria's Mobile-Payments Market

The situation

Between 2016 and 2025, the share of payment value handled by banks' own apps and USSD fell from **78%** to **28%**. Fintech platforms — agents, POS terminals and apps — took the rest. One platform, 'KoboLink', now handles **30%**.

The decision question

Is a dominant payments platform **good for customers** (scale, convenience, low cost) or a **competition problem** (lock-in, higher fees, blocked rivals)?

The data we have

payments_market_shares.csv — 10 years × 6 providers: share of payment value, active users, merchants and average fee. *Stylised; providers other than the banks are fictional.*

What a good answer looks like

A **reasoned view for the regulator**: how concentrated the market is, whether fees signal market power, and what — if anything — should be done.

PART

01

Market Structure

How many rivals, and how different are they?

THE SPECTRUM

Four Market Structures

Structure	Sellers & products	Price power	Nigerian example
Perfect competition	Many sellers, identical products, easy entry	None — price taker	Tomatoes at Mile 12 market, foreign-exchange street traders
Monopolistic competition	Many sellers, differentiated products	Some — brand loyalty	Restaurants, fashion, hair salons, noodles brands
Oligopoly	A few large sellers; each watches the others	Significant — strategic	Cement, telecoms, breweries, large banks
Monopoly	One seller, no close substitute, high barriers	High — price maker	Electricity distribution in a franchise area, national grid transmission

*Most real markets sit **between** these boxes. The question is always: how much can a firm raise price before customers leave?*

What Market Power Costs Society

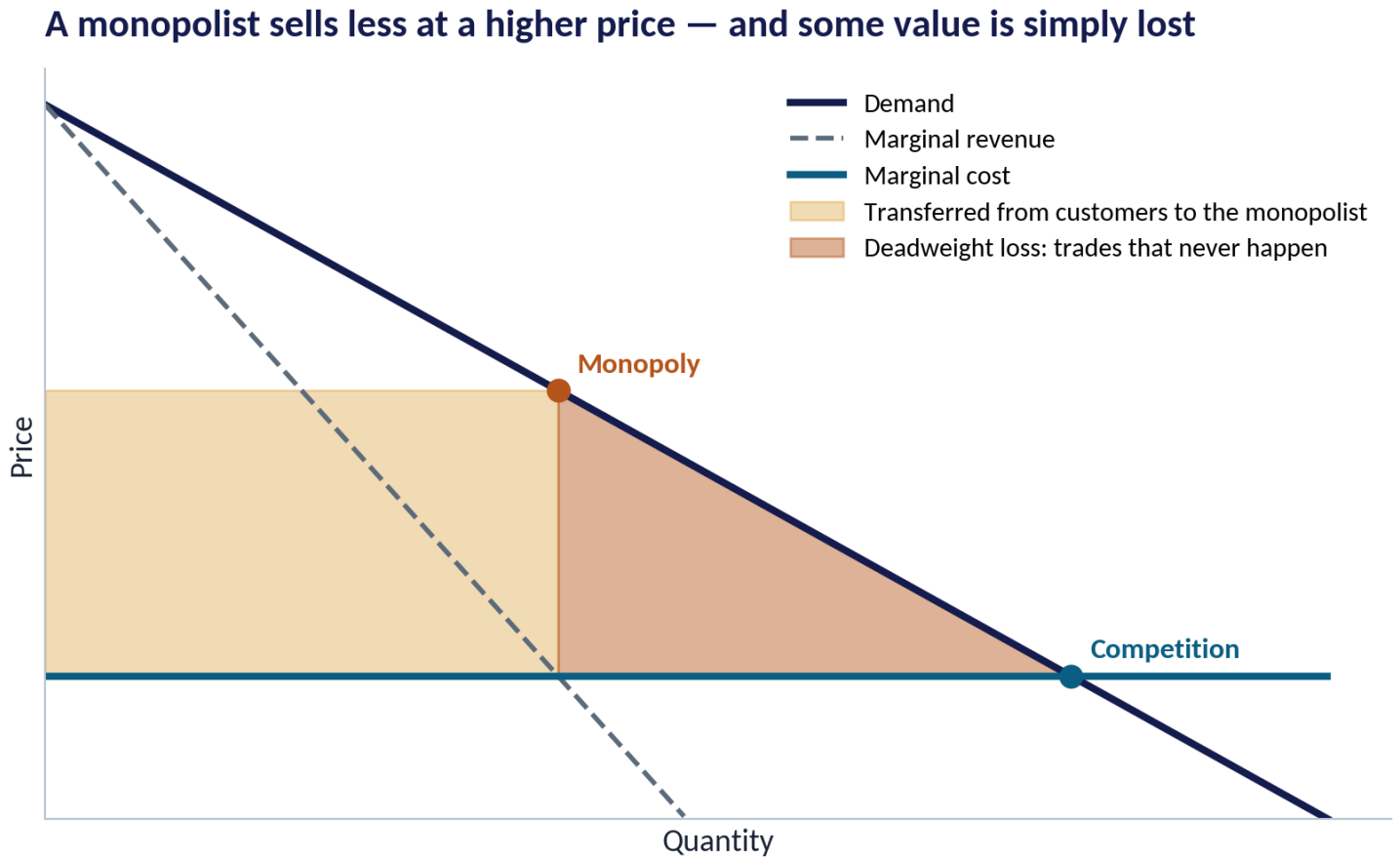


Fig S04-F06 · replicate: s04_figures.R / .py

WHAT TO SEE

- Under competition, price falls to the cost of the last unit, and output is high.
- A monopolist restricts output to push the price up.
- Gold area: value **transferred** from customers to the monopolist.
- Red area: **deadweight loss** — purchases worth more than they cost that simply never happen.

The Herfindahl–Hirschman Index (HHI)

IN PLAIN ENGLISH

The HHI adds up the **squared market shares** of every firm. Squaring gives big firms more weight. A pure monopoly scores **10,000**; many tiny firms score close to **0**.

EXAMPLE

Four firms with 40%, 30%, 20%, 10%: 1,600 + 900 + 400 + 100 = **3,000** — highly concentrated.

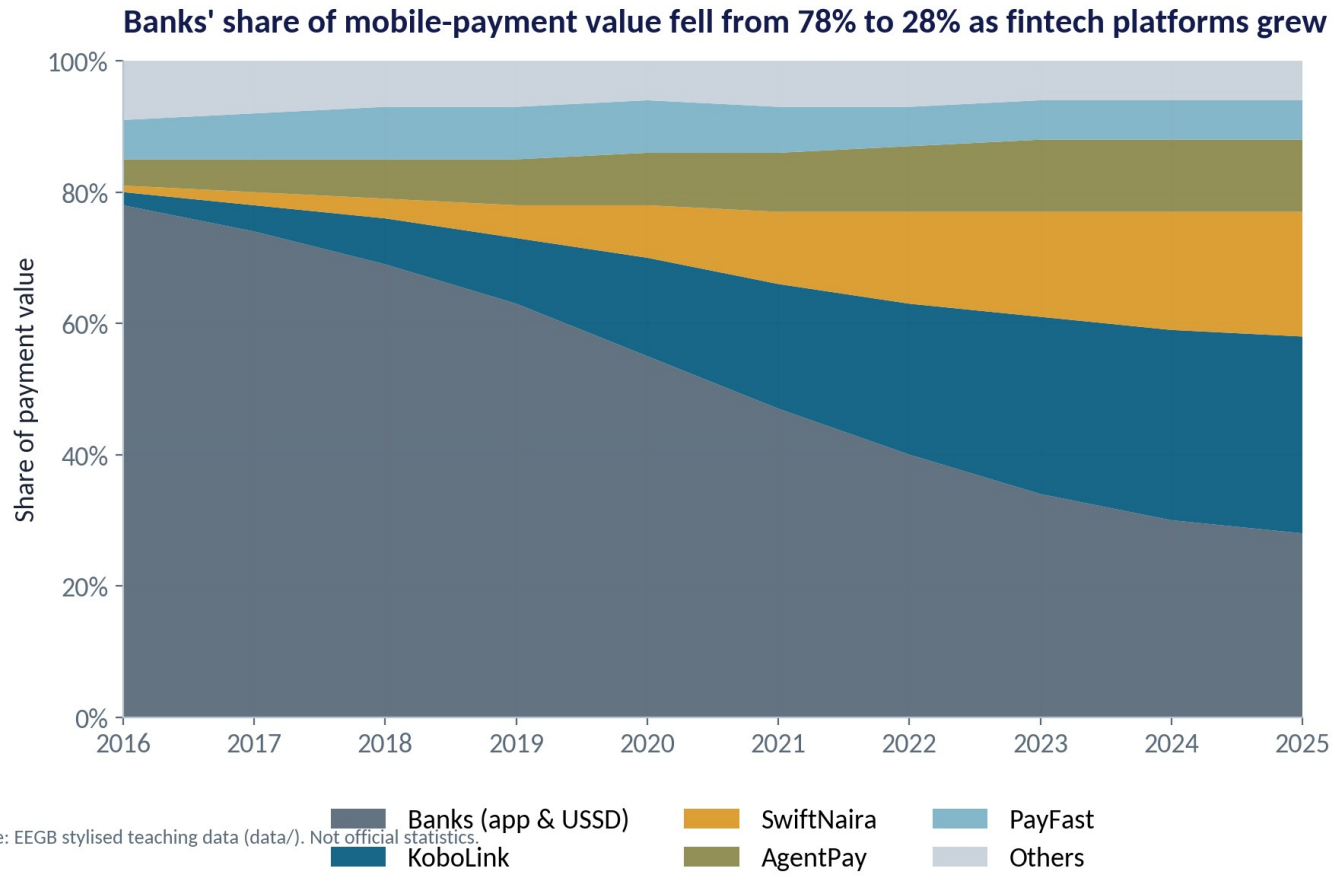
THE FORMULA

$$HHI = s_1^2 + s_2^2 + \ldots + s_n^2 \quad \text{(shares in \%)}$$

WHY A MANAGER CARES

Competition authorities use the HHI to screen mergers. A merger's effect on the index is roughly $2 \times s_1 \times s_2$ — two firms of 15% and 10% add 300 points.

From Banks to Platforms



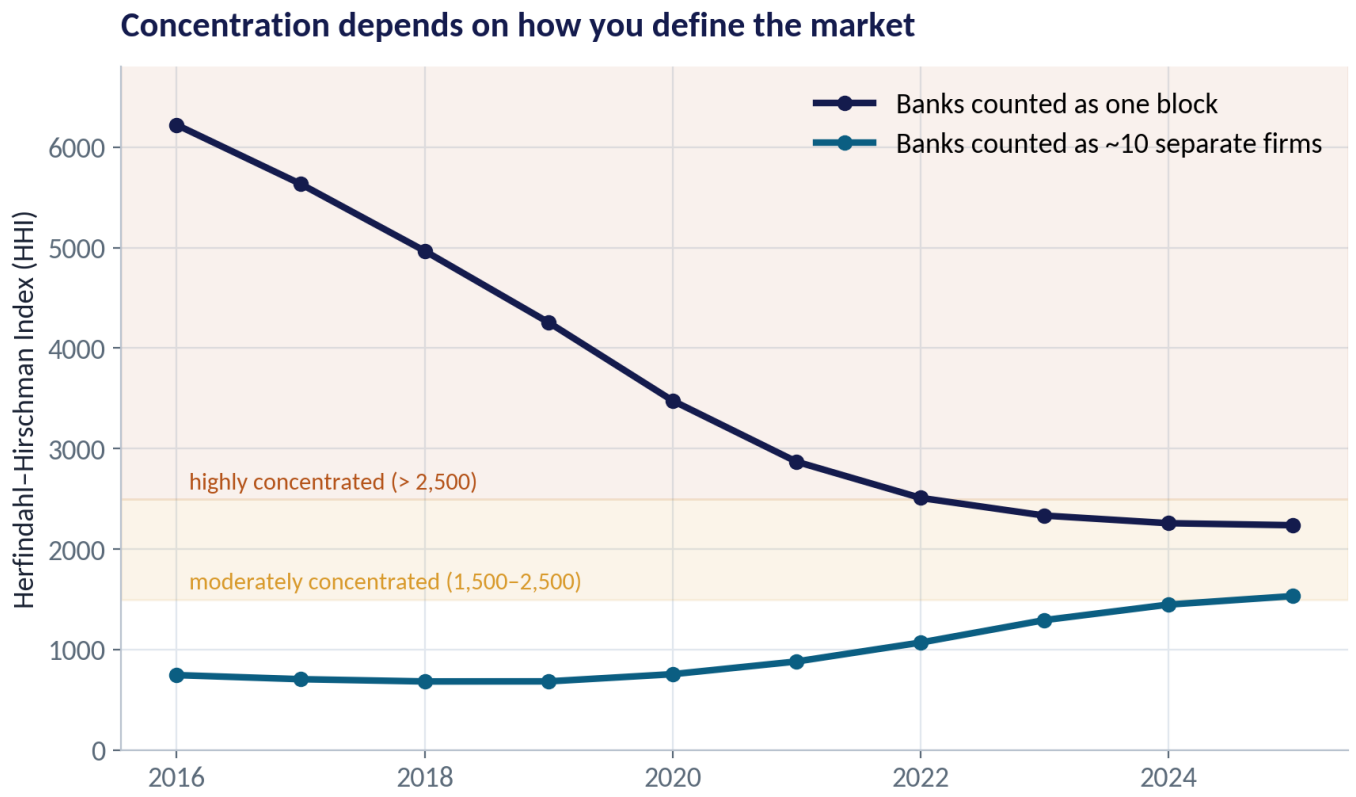
Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S04-F01 · replicate: s04_figures.R / .py

WHAT TO SEE

- In 2016 banks' apps and USSD handled **78%** of mobile-payment value.
- By 2025: banks **28%**, KoboLink **30%**, SwiftNaira **19%**, AgentPay **11%**.
- Fintechs won by going where banks were weak: agents in markets, cheap POS, simple apps.
- The market became **more competitive** between banks and fintechs — and **more concentrated** among fintechs.

Concentration Depends on How You Define the Market



Thresholds: US 2010 merger guidelines (the 2023 guidelines use lower cut-offs). Source: EEGB stylised teaching data.

Fig S04-F02 · replicate: s04_figures.R / .py

WHAT TO SEE

- Treat 'banks' as one block: HHI falls from **6,222** to **2,238** — the market looks ever more competitive.
- Treat them as ~10 separate banks: HHI **rises** from **746** to **1,532** — concentration is growing.
- Same data, opposite stories. **Defining the market** is the first and most contested step in any competition case.
- Ask: who do customers actually switch to when one provider raises its price?

Quick Check: What Would You Say?

?

A market has five firms with shares 30%, 25%, 20%, 15% and 10%. What is the HHI?

- **A** 100
- **B** 2,250
- **C** 1,000
- **D** 10,000

Discuss with your neighbour for 60 seconds — answer and reasoning in the speaker notes.

PART

02

Platforms and Network Effects

Why digital markets tip towards one winner.

A Platform Serves Two Groups at Once

IN PLAIN ENGLISH

A **platform** connects two groups who need each other: payers and merchants, riders and drivers, buyers and sellers. Each side values the platform **more when the other side is bigger**.

EXAMPLE

A trader accepts KoboLink because many customers pay with it; customers download it because many traders accept it.

THE FORMULA

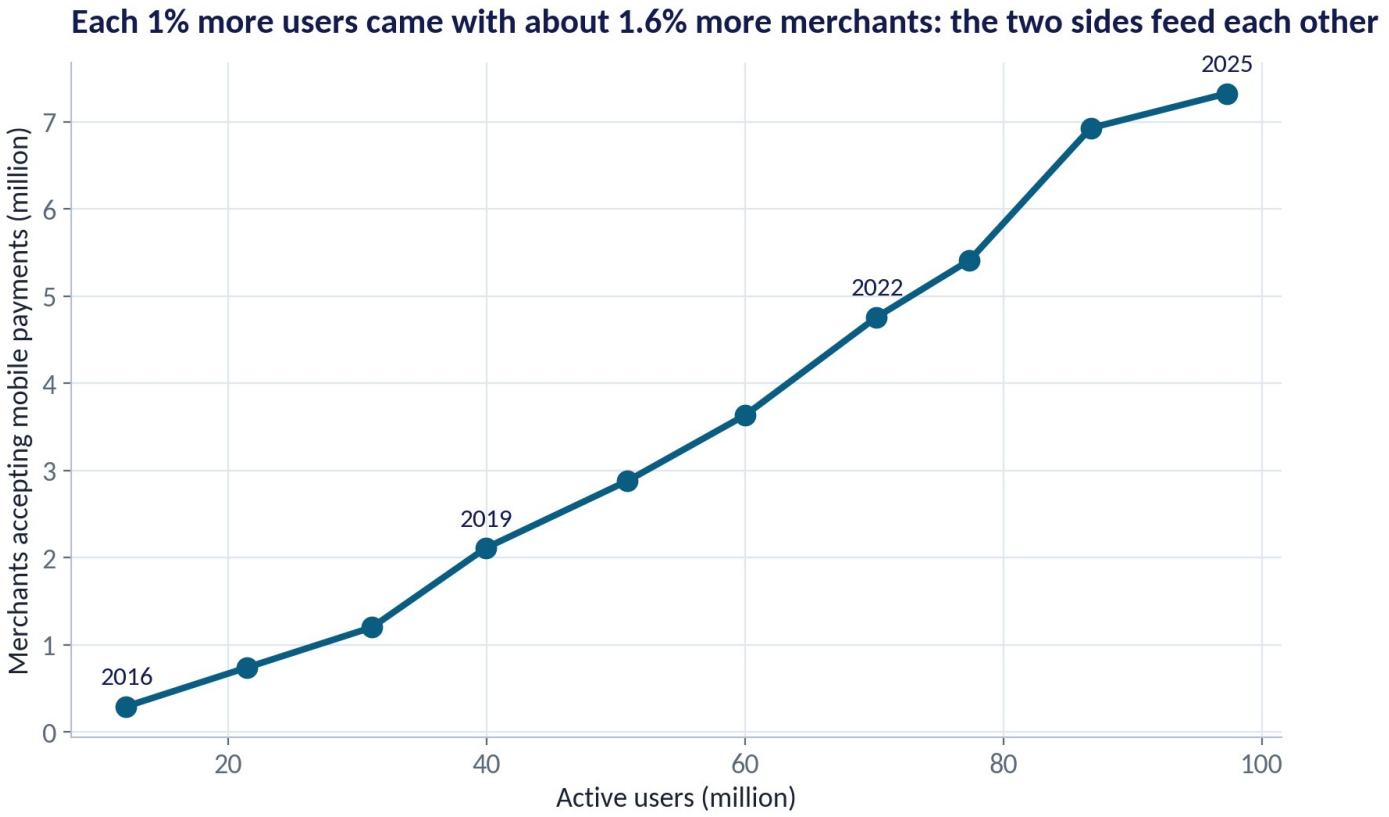
value to a user ↑ as merchants ↑

and value to a merchant ↑ as users ↑

WHY A MANAGER CARES

Platforms often **subsidise one side** (free app for users, free POS for agents) to win the other. Low prices today can be a strategy, not a gift.

Users and Merchants Grew Together



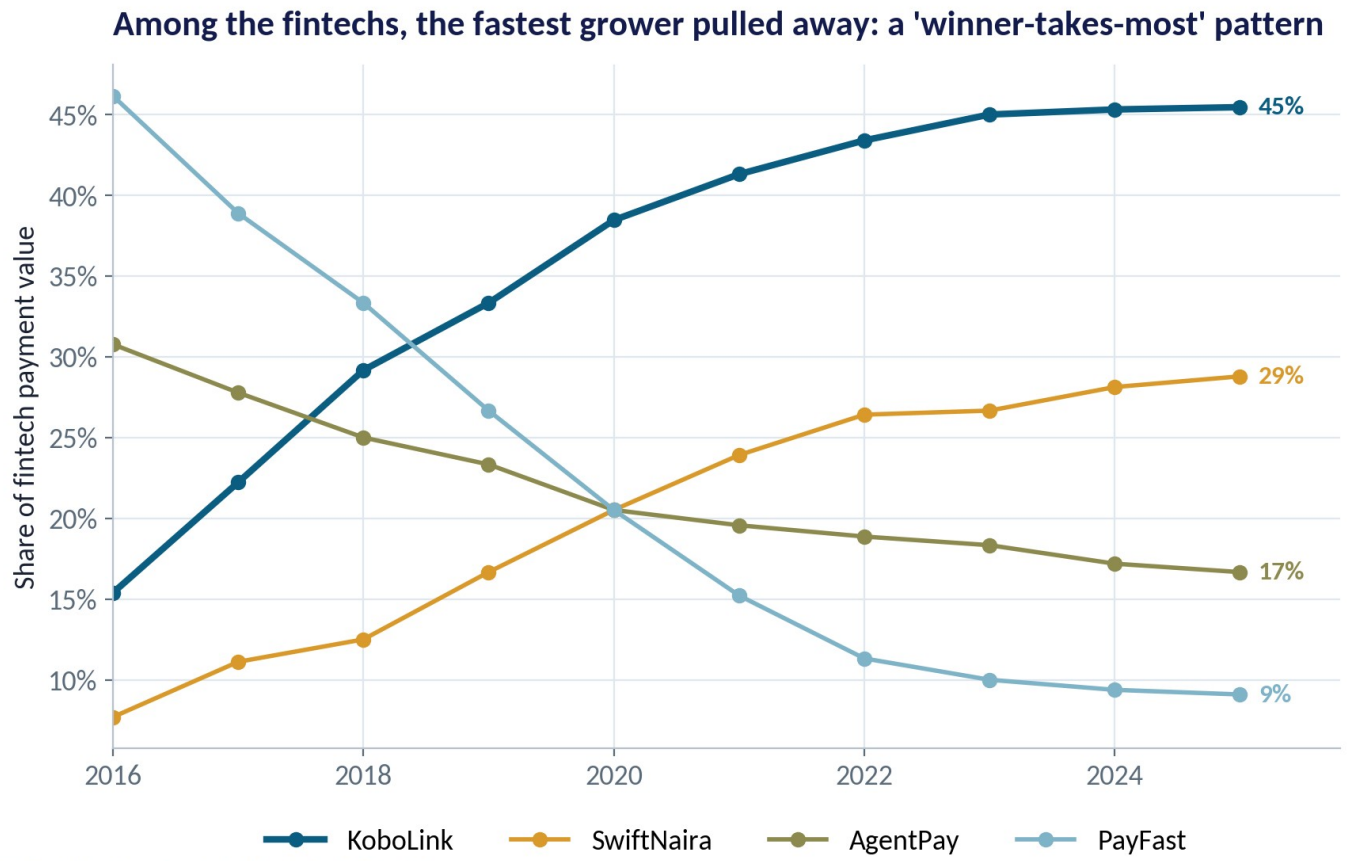
Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S04-F03 · replicate: s04_figures.R / .py

WHAT TO SEE

- Active users rose from **12 million to 97 million**; merchants from **0.3 million to 7.3 million**.
- Each 1% more users came with about **1.6%** more merchants.
- The two sides **reinforce** each other: more users attract merchants, who attract more users.
- This feedback loop is why platform markets can grow very fast — and **tip** towards a leader.

Winner-Takes-Most Among the Fintechs



Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S04-F04 · replicate: s04_figures.R / .py

WHAT TO SEE

- Within the fintech segment, KoboLink went from **15%** to **45%** of payment value.
- PayFast, the early leader, fell from **46%** to **9%**.
- Being first is not enough: the platform that builds the **network fastest** wins.
- Once a network tips, rivals need a big advantage — or regulation — to catch up.

Why Big Platforms Are Hard to Challenge

Network effects

Everyone is on the leading platform because everyone else is. A new entrant starts with an empty network.

Switching costs

Moving your shop's payments, savings history or credit record to a new app takes time and risk. Customers stay even when a rival is a bit cheaper.

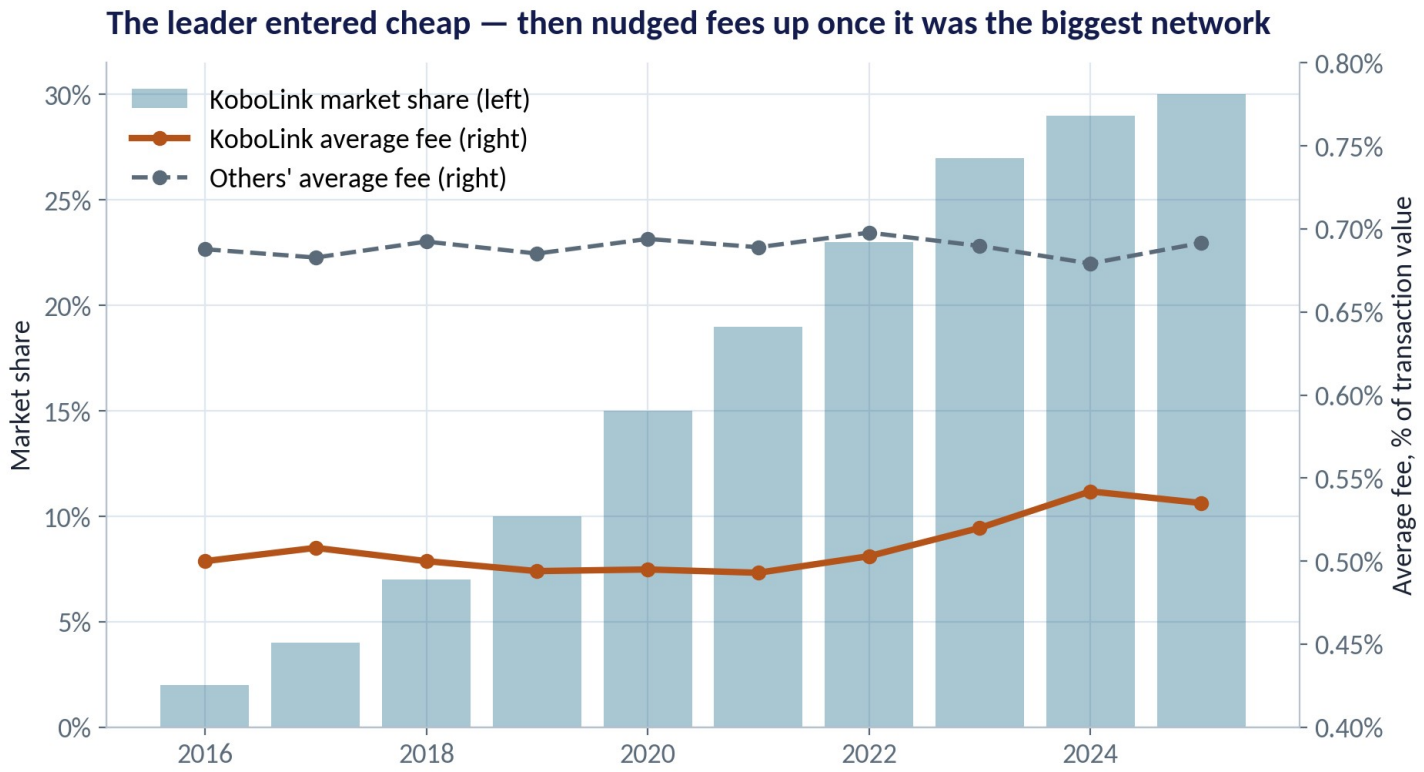
Data

Transaction data lets the leader score credit, spot fraud and target offers better than rivals — and the advantage grows with size.

Bundling & ecosystems

Payments + loans + savings + bills in one app. Each service makes leaving the others harder.

Fees After Dominance



Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S04-F05 · replicate: s04_figures.R / .py

WHAT TO SEE

- KoboLink entered **cheaper** than rivals: about **0.50%** of value, against roughly **0.69%** for others.
- As its share passed a quarter of the market, its fee rose to about **0.54%**.
- Still below rivals — so customers are not obviously harmed yet.
- A regulator would watch the **trend**, merchant complaints, and whether rivals can still win business.

When a Platform Becomes a Bank

IN PLAIN ENGLISH

Ant Financial (now Ant Group) grew from Alipay, a payments tool, into lending, wealth management and insurance for hundreds of millions of Chinese users — using payments data to score credit.

EXAMPLE

In 2020 Chinese regulators halted Ant's planned stock-market listing and required it to restructure under bank-style rules.

THE FORMULA

payments → data → credit → more payments

WHY A MANAGER CARES

Nigerian fintechs follow a similar path from payments to lending. The case asks: should a platform that behaves like a bank be **regulated like a bank**?

PART

03

Competition Policy

When does big become bad?

TWO VIEWS OF A DOMINANT PLATFORM

Efficiency or Exclusion?

Good for customers

Scale lowers costs and adds convenience

- One network everyone can use
- Lower fees from economies of scale
- Fast innovation and new services
- Financial inclusion for the unbanked

A competition problem

Power used to block rivals and raise prices

- Exclusive deals with agents and merchants
- Rising fees once rivals are weakened
- Self-preferencing its own loans and products
- Data hoarding that rivals cannot match

*Nigeria's **FCCPC** (under the 2018 competition act) and the **CBN** (for payments) share the job. The question is conduct, not size.*

What Authorities Can Do

Tool	What it does	Example in payments
Merger review	Blocks or conditions deals that raise concentration too much	Screen mergers that add more than ~200 HHI points in a concentrated market
Abuse-of-dominance cases	Punishes exclusionary conduct by a dominant firm	Ban exclusive agent contracts that forbid rival POS terminals
Interoperability rules	Makes networks connect to each other	Require transfers between all wallets and banks at fair fees
Data portability	Lets customers take their data to a rival	Open banking: share transaction history with a new lender
Price or fee caps	Limits charges directly — a last resort	Caps on transfer or withdrawal fees

*Interoperability and data portability weaken network effects **without** breaking up a successful firm.*

Common Mistakes About Competition

THE TRAP	THE BETTER HABIT
✗ “They have 30% of the market, so they must be abusing it.”	✓ Size is not abuse. Look at **conduct**, entry and price trends.
✗ “Fees are lower than rivals', so there is no problem.”	✓ Low prices can be **penetration pricing**; watch what happens after rivals weaken.
✗ Measuring concentration without defining the market.	✓ Start with **who customers switch to**; test more than one definition.
✗ Treating free products as harmless.	✓ When the product is free, users often pay in **data or lock-in**.
✗ Assuming network effects always end in monopoly.	✓ **Interoperability, multi-homing** and new technology can keep markets open.

CONCEPT CHECK

Quick Check: What Would You Say?

?

SwiftNaira (19%) and AgentPay (11%) want to merge. Using HHI ≈ 1,532 before the deal, what does the merger do?

A Raises the HHI by about 30 points — no concern

B Raises the HHI by about 418 points, taking it to about 1,950 — the deal deserves close review

C Lowers the HHI because the merged firm can compete with KoboLink

D Has no effect: HHI only changes when firms enter or exit

Discuss with your neighbour for 60 seconds — answer and reasoning in the speaker notes.

HANDS-ON LAB

Lab: Is the Payments Market Competitive?

YOU WILL NEED

Data
`data/payments_market_shares.csv`

R script
`sessions/s04/R/s04_lab.R`

Python script
`sessions/s04/python/s04_lab.py`

⏱ **20 minutes**

1. Tabulate market **shares** by provider and year.
2. Compute **CR2** and the **HHI** for each year.
3. Recompute the HHI treating 'Banks' as **ten** separate banks — compare the two stories.
4. Measure two-sided growth: % more merchants per 1% more users.
5. Check whether the leader's **fee** rises with its share; test the SwiftNaira–AgentPay **merger**.

Deliverable: a 3-sentence note to the competition regulator on the proposed merger.

WORDS TO OWN

Today's Vocabulary in Plain English

Market power	Ability to raise price above cost without losing most customers.
Oligopoly	A market dominated by a few large, interdependent sellers.
Deadweight loss	Value lost because worthwhile trades do not happen.
Concentration ratio (CR)	Combined share of the largest firms, e.g. CR2.
HHI	Sum of squared market shares; 0 to 10,000.
Two-sided market	A platform serving two groups who value each other.
Network effect	A product becomes more valuable as more people use it.
Switching cost	The cost or hassle of moving to a rival.
Tipping	When a network market swings decisively to one leader.
Interoperability	Rules that let rival networks connect and exchange.

KEY TAKEAWAYS

Session 4 in 5 Ideas

01 Market structure runs from **perfect competition to monopoly**; most markets sit in between.

02 Market power **raises prices, cuts output** and creates deadweight loss.

03 Measure concentration with **CR and HHI** — but **define the market** first.

04 Platforms grow through **network effects**, switching costs and data — and can **tip**.

05 Regulators judge **conduct, not size**; interoperability keeps markets open.



NEXT › Session 5: Market Failures

Before next session: Read 'Externalities: Justification for Public Goods?' (210-023-1) and the UVA note on pollution taxes and cap-and-trade. Think about one cost your industry imposes on others that it does not pay for.